**Project Design Phase**

**Problem – Solution Fit Template**

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| Date | 12-10-2026 |
| Team ID | SWUID2026403484 |
| Project Name | HotelPro - Hotel Room Booking and Management System |
| Maximum Marks | 2 Marks |

## Problem – Solution Fit Overview:

The Problem–Solution Fit ensures that the HotelPro platform addresses real gaps in how people discover, reserve and manage hotel accommodations, and aligns with the needs of customers, hotel staff/receptionists and administrators. This validation is critical before the system is built and scaled.

## Purpose:

- Bridge the gap between customers who want a simple, fast and secure online platform to find and book hotel rooms, and hotel staff and administrators who want to manage properties and operations efficiently.
- Offer one unified full-stack web platform covering the entire hospitality journey, from room search and real-time availability checking to online booking, payment processing and front-desk check-in/check-out.
- Eliminate double bookings and payment confusion by maintaining accurate real-time room availability, re-verifying availability at confirmation and ensuring bookings are confirmed only after successful payment.
- Reduce manual paperwork and effort for hotel staff and administrators with digital tools to manage hotels, room inventory, guest records, reservations and billing from a single system.
- Provide administrators with actionable insights through a dedicated dashboard displaying bookings, occupancy rates and revenue metrics, backed by a modular MERN stack and JWT-based role-based access control.

## Problem Statement:

Many customers, hotel front-desk staff and property administrators face challenges such as:

- Difficulty searching and filtering available rooms across hotels by city, dates, guest capacity, price, room type and amenities in one place
- Two customers attempting to reserve the same room for overlapping dates, leading to double bookings and conflicting reservations

- Manual paperwork and time-consuming front-desk operations during guest registration, check-in, check-out and room status tracking
- Unclear booking modification and cancellation processes with poor visibility into reservation and payment refund statuses
- Fragmented tracking of primary and accompanying guest details, contact information, ID proofs and stay histories
- Limited administrative tools to monitor real-time occupancy rates, active bookings, cancellations and revenue performance across properties

## Solution:

**HotelPro**, a full-stack hotel room booking and management system, offers:

- City and hotel-based room discovery with real-time search by check-in/check-out dates and guest count, plus filtering and sorting by price, room type, amenities and rating
- Real-time availability verification that displays only unbooked rooms for the selected date range and re-verifies availability upon confirmation to prevent double booking
- End-to-end room booking flow with automatic price calculation (nightly rate multiplied by nights plus taxes), guest details capture, unique booking ID generation and confirmation summary
- Secure online payment processing (credit/debit card, UPI, net banking) via an integrated payment gateway plus offline counter payment recording, automated invoicing and status tracking
- Comprehensive booking management allowing customers to view, modify or cancel reservations before check-in (with refund support and automatic room release) and staff to filter all bookings
- Digital guest management and front-desk check-in/check-out workflows that record ID proofs, accompanying guests and actual timestamps while automatically updating room statuses (Available, Occupied, Maintenance)
- Centralized hotel and room inventory management enabling administrators to add, update or deactivate hotels and rooms (Single, Double, Deluxe, Suite) while protecting rooms with active bookings from deletion
- Role-based portals for Customers, Hotel Staff and Administrators secured with JWT authentication and bcrypt password hashing, featuring an Admin Dashboard with revenue/occupancy analytics and CSV/PDF report exports